Session 2 — Paper exercises

Reference: Sutton & Barto, *Reinforcement Learning: An Introduction* (2nd ed.), Chapters 4 and 6. Python code for this gridworld (Shangtong Zhang). No computer needed.

The 4×4 gridworld (Example 4.1)

States are numbered as in the book. The two shaded corners are the same terminal state, written T:

T	1	2	3
4	5	6	7
8	9	10	11
12	13	14	T

- Actions: up, down, left, right. Moves are **deterministic**; an action that would leave the grid leaves the state unchanged.
- Reward: -1 for every transition, until the terminal state is reached.
- Policy: equiprobable random policy, $\pi(a \mid s) = 1/4$ for every action.
- No discounting: $\gamma = 1$.

Because the transitions are deterministic, action a in state s always leads to the same next state s'_a with the same reward r_a. The Bellman expectation equations become:

$$q_\pi(s,a) = r_a + \gamma\, v_\pi(s'_a), \qquad v_\pi(s) = \sum_a \pi(a \mid s)\, q_\pi(s,a) = \sum_a \pi(a \mid s)\big(r_a + \gamma\, v_\pi(s'_a)\big).$$

The value function of the random policy (from the lecture) is:

0	-14	-20	-22
-14	-18	-20	-20
-20	-20	-18	-14
-22	-20	-14	0

Part A — Values and policy improvement

A1 (Exercise 4.1). Compute $q_\pi(11, \text{down})$.
Answer:

A2 (Exercise 4.1). Compute $q_\pi(7, \text{down})$.
Answer:

A3. Draw the **greedy policy** with respect to v_π: in each state, the action(s) leading to the neighbour with the highest value.

1. What do you notice about this policy?
2. The policy improvement theorem says that the greedy policy π' satisfies $v_{\pi'} \geq v_\pi$. Does the theorem guarantee that π' is *optimal*? What happens if we apply policy improvement again?

Answer:

Part B — Connecting mathematics and code

The following line comes from an implementation of iterative policy evaluation for this gridworld (`state_values` holds the current estimate V):

```
value += ACTION_PROB * (reward + discount * state_values[next_i, next_j])
```

B1. Which part of the Bellman expectation equation does this line implement? Why is there no sum over s'?
Answer:

Part C — The AB example (Sutton & Barto, Example 6.4)

We observe 8 episodes in a Markov reward process with two states A and B, no discounting:

episode 1:	A, 0, B, 0
episodes 2–7:	B, 1 (six times)
episode 8:	B, 0

("A, 0, B, 0" means: start in A, reward 0, move to B, reward 0, end.)

C1. What estimates $V(A)$ and $V(B)$ does **Monte Carlo** give (average of the observed returns)?
Answer:

C2. Now replay these 8 episodes over and over with **TD(0)** until the values stop changing. What are $V(A)$ and $V(B)$? *Hint: TD uses the update target* $R + V(S')$.
Answer:

C3. Which answer is "right"? Explain what each method is actually computing.
Answer:

Final reflection

In **one sentence**, explain the difference between dynamic programming and the learning methods of today (MC, TD).
Answer: